KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

College of Engineering

ME 365: Thermodynamics II

SECOND SAMPLE PROBLEMS

BRAYTON CYCLE

Answers are rounded up. They may even contain errors. Please report any error.

k = 1.4; M = 29; R = 8.314 kJ/kmol K; cp = 1.005 kJ/kg.K

1. Air enters the compressor of an ideal air standard Brayton cycle at 100 kPa, 25oC, with a volumetric flow rate (V·) of 8 m3/s. The compressor pressure ratio is 12. The turbine inlet temperature is 1100oC. Determine (a) the thermal efficiency (ηth), (b) net power output (W·net) and (c) back work ratio.

Ans: (a) 50.83 %, (b) 3666.08 kW, (c) 44.15%

2. A stationary power plant operating on an ideal Brayton cycle has a pressure ratio of 7. The gas temperature is 25°C at the compressor inlet and 1000°C at the turbine inlet. Utilizing the air standard assumptions, determine (a) the gas temperature (*T*) at the exits of the compressor, (b) back work ratio and (c) the thermal efficiency ($\eta_{th,Brayton}$).

Ans: (a) 519.6 K (b) 40.82%, (c) 42.65%

3. A stationary power plant operating on a simple Brayton cycle has a pressure ratio of 10. The compressor efficiency is 80% and its inlet conditions are 100 kPa and 300 K with a flow rate of 10 m3/s. The turbine efficiency is 85% and its inlet temperature is 1400 K. Using the air standard Brayton cycle to model the power plant, determine (a) the power output (W·net), (b) the thermal efficiency (ηth), and (c) the back work ratio. (d) What would the thermal efficiency and net power be if the compressor efficiency dropped to 70%?

Ans: (a) 2622.94 kW (b) 29.91 %, (c) 60.84 % , (d) 24.93%, 2040.76 kW

4. Air enters the compressor of a regenerative gas-turbine engine at 100 kPa and 300 K and is compressed to 950 kPa. The regenerator has an effectiveness of 80 percent, and the air enters the turbine at 1200 K. For a compressor efficiency of 80 percent and a turbine efficiency of 85 percent, determine

(a) The heat transfer in the regenerator

(b) The back work ratio

(c) The cycle thermal efficiency

Ans: a) 62.41 kJ/kg, (b) 69.95 %, (c) 29.12 %

VAPOUR COMPRESSION REFRIGERATION CYCLE

5. Refrigerant-134a is the working fluid in an ideal compression refrigeration cycle. The refrigerant leaves the evaporator saturated at -24oC and has a condenser pressure of 1 MPa. The mass flow rate is 0.1 kg/s. Show the cycle on a P-h diagram with respect to saturation lines. Determine (a) the rate of heat removal from the refrigerated space (b) the power input to the compressor, (c) the rate of heat rejection to the environment, and (d) the coefficient of performance, (e) what is the COP of the equivalent heat pump.

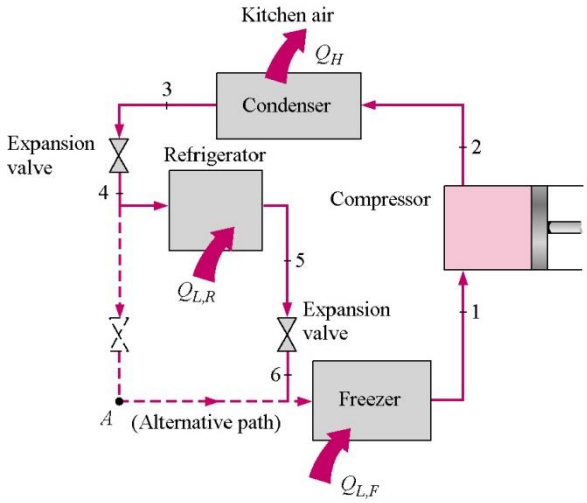
Answers: (a) 13.13 kW (b) 4.55 kW, (c) 17.88 kW, (d) 2.93, (e) 3.93

6. A refrigerator uses R-134a as the working fluid and operates on an ideal vapor-compression refrigeration cycle between 0.18 MPa and 1.2 MPa. Assume that the refrigerant leaves the evaporator saturated. Show the cycle on a T-s diagram with respect to saturation lines. For a cooling load of 15 kW, determine (a) the heat removed from the refrigerated space (b) the energy input to the compressor, (c) the COP and (d) the mass flow rate of the refrigerant through the evaporator.

Answers: (a) 123.95 kJ/kg, (b) 39.39 kJ/kg, (c) 3.15, (d) 0.121

7. The refrigerator-freezer unit, shown in the schematic below, uses R-134a as the working fluid and operates on an ideal vapor-compression cycle. The condenser pressure is 0.5 MPa. The refrigerant leaves the refrigerator, and the freezer at 2°C, and -20°C, respectively. The mass flow rate of the refrigerant is 0.1 kg/s. Show the cycle on a T-s diagram with respect to saturation lines. If the refrigerant quality at the refrigerator exit is 0.4, determine the rate of heat removal from (a) the refrigerator and (b) freezer. Also, determine (c) the compressor power input and (d) the COP of the unit.

Answers: (a) 2.77 kW, (b) 10.23 kW, (c) 4.18 kW, (d) 3.11



NON-REACTING IDEAL GAS MIXTURES

AIR-VAPOUR MIXTURES AND AIR CONDITIONING

8. The volumetric analysis of an ideal gas mixture shows 74 % Nitrogen (N_2), 7 % Oxygen (O_2), 8 % Carbone dioxide (CO_2) and 11% of water vapour (H_2O).

a) Determine the apparent molecular weight of the mixture

b) Find the analysis on a mass basis

b) What is the mass of 3 m^3 of this gas when P = 2 MPa and T = 450 K?

c) Find the specific heats (C_{pm} and C_{vm}) of the mixture at 450 K

d) This gas is heated in a steady state steady flow process such that its temperature is increased by 150 K. Find the required heat transfer

e) This mixture undergoes an isentropic process from 2 MPa, 300 K to 0.2 MPa. Find T2

(a) 28.48 kg/kmol

(b)

	Mf
N2	0.7280
CO2	0.1237
O2	0.0787
H2O	0.0696

(c) 68.46 kg; (d) 1.093 kJ/kg. K, 0.801 kJ/kg. K; (e) 218.78 kJ/kg; (f) 181.45 K

9. The air at 35°C and 100 kPa in a 200 m^3 room has a relative humidity of 60%.

Calculate:

a) the humidity ratio

b) the dew point

c) the mass of water vapor in the air

d) the volume percentage of the room that is water vapor

The air is cooled below the dew point to 10°C.

e) Estimate the percentage of water vapor that will condense.

Answers: (a) 0.021738.77 kgv/kga, (b) 25.92oC, (c) 4.75 kg, (d) 3.38% (e) 64.57%

COMBUSTION

10. Methane is burned with the theoretical amount of air during a combustion process. Assuming complete combustion, determine the air-fuel ratio on (a) mass basis, (b) mole basis and (c) volume basis. **Answers: (a) 17.17, (b) 9.52, (c) 9.52**

11. For a theoretical (stoichiometric) hydrogen-air reaction at 200 kPa, find (a) the fuel-to-air mass ratio, (b) the mass of fuel per unit mass of reactants, (c) the partial pressure of water vapor in the products. Assume products temperature to be high enough for all H_2O to be in vapor form. (d) What is the minimum temperature of the products to ensure that water stays as vapor in the products.

12. Ethane (C_2H_6) is burned with 30% excess air during a combustion process. Assuming complete combustion and a total pressure of 100 kPa, determine (a) the air-fuel ratio and (b) the dew point temperature of the products. (c) What would the air to fuel ratio be if methane were burned instead of ethane?

13. Determine the mass-based air-fuel ratio for the complete combustion of methane, CH4, if the air supplied is (a) theoretical amount, (b) 150% theoretical air. What would the theoretical amount be if the fuel were (c) octane, (d) dodecane?

Answers: (a) 17.26 kg air/kg fuel (b) 25.89 kg air/kg fuel (c) 15.09 kg air/kg fuel (d) 15.01 kg air/kg fuel

Please, enjoy!!!

LECTURER: E.W. RAMDE